

# Redundant Power Selector Malfunction Low Voltage

## Symptoms

- Primary power supply failure error message.

## How To Use This Tree

This symptom tree can be used to troubleshoot signal from the speed pick-up is signal lost. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

## Shoptalk

Primary power supply error message.

## Troubleshooting Summary

| STEPS   |                                                                                                | SPECIFICATIONS                    |
|---------|------------------------------------------------------------------------------------------------|-----------------------------------|
| STEP 1. | Check the customer interface box (C.I.B.) wiring.                                              |                                   |
|         | STEP 1A. Check the control panel display for faults.                                           | Control panel indicates fault(s)? |
| STEP 2. | Check redundant power selector wiring.                                                         |                                   |
|         | STEP 2A. Check redundant power selector primary power input SUPPLY wire for voltage +18 VDC.   | Less than +18±0.2-VDC?            |
|         | STEP 2B. Check redundant power selector secondary power input SUPPLY wire for voltage +18 VDC. | Less than +18±0.2 VDC?            |
| STEP 3. | Check redundant power selector voltage.                                                        |                                   |
|         | STEP 3A. Check redundant power selector power output SUPPLY.                                   | Output within ± 0.5 VDC?          |

## STEP 1. Check the customer interface box (C.I.B.) wiring.

### STEP 1A. Check the control panel display for faults.

**Conditions :**

- Locate the control panel display.

| Action                                      | Specification/Repair                                                                                                                                                                                                                                                                                                                                                                                                            | Next Step                                                 |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Check the control panel display for faults. | Control panel indicates fault(s)?<br><br><b>YES</b><br><br><b>Repair :</b><br><br>Troubleshoot the appropriate fault code. Reference the Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF, or the ISM and QSM 11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF, X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346. | <b>Go to appropriate fault code troubleshooting tree.</b> |
|                                             | Control panel indicates fault(s)?<br><br><b>NO</b><br><br><b>Repair :</b><br><br>Replace the drive coupling.                                                                                                                                                                                                                                                                                                                    | <b>2A</b>                                                 |

## STEP 2. Check redundant power selector wiring.

### STEP 2A. Check redundant power selector primary power input SUPPLY wire for voltage $+18 \pm 0.2$ VDC.

**Conditions :**

- Open the C.I.B.
- Test the redundant power selector primary power input SUPPLY wire pin 1.

| Action                                                                                                                                                                                                                                                                                                                                                            | Specification/Repair                                                                                                                    | Next Step |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Check the voltage at pin 1 of the redundant power selector. <ul style="list-style-type: none"><li>• Place one test lead at the primary power SUPPLY wire at pin 1 of the redundant power selector.</li><li>• Place the other test lead on the panel ground.</li></ul> Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than +18±0.2 VDC?<br><b>YES</b><br><br><b>Repair :</b><br><br>Check the batteries. See equipment manufacturer service information. | <b>2B</b> |
|                                                                                                                                                                                                                                                                                                                                                                   | Less than +18±0.2 VDC?<br><b>NO</b>                                                                                                     | <b>2B</b> |

**STEP 2B. Check redundant power selector secondary power input SUPPLY wire for voltage +18±0.2 VDC.****Conditions :**

- Open the C.I.B.
- Test the redundant power selector secondary power input SUPPLY wire.

| Action | Specification/Repair | Next Step |
|--------|----------------------|-----------|
|--------|----------------------|-----------|

|                                                                                                                                                                                                                                                                                                                                                                                  |                                                 |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------|
| <p>Check the voltage at pin 6 of the redundant power selector.</p> <ul style="list-style-type: none"> <li>Place one test lead at the secondary power SUPPLY wire at pin 6 of the redundant power selector.</li> <li>Place the other test lead on the panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than +18±0.2 VDC?</p> <p><b>YES</b></p> | <p><b>Repair complete.</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                  | <p>Less than +18±0.2 VDC?</p> <p><b>NO</b></p>  | <p><b>3A</b></p>               |

### STEP 3. Check redundant power selector power output SUPPLY.

#### STEP 3A. Check redundant power selector power output wire for voltage.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the C.I.B.</li> <li>Test the redundant power selector power output SUPPLY wire.</li> </ul> |                      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| Action                                                                                                                                                            | Specification/Repair | Next Step |

|                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                           |                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <p>Check the output voltage at pin 17 of the redundant power selector.</p> <ul style="list-style-type: none"> <li>Place one test lead at the power output wire at pin 17 of the redundant power selector.</li> <li>Place the other test lead on the panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Secondary power input SUPPLY equals the redundant power selector output <math>\pm 0.5</math> VDC?</p> <p><b>YES</b></p>                                                                | <p><b>Repair complete.</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                 | <p>Secondary power input SUPPLY equals the redundant power selector output <math>\pm 0.5</math> VDC?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace redundant power selector.</p> | <p><b>1A</b></p>               |